Determining Best EEZ's for Oysters

EDS 223 - HW 3

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Prepare Data

Load packages

```
library(sf)
library(terra)
library(tidyverse)
library(tmap)
```

```
library(elevatr)
library(tigris)
library(viridisLite)
library(glue)
```

```
# West Coast EEZ
EEZ <- st_read("data/wc_regions_clean.shp")
```

```
Reading layer `wc_regions_clean' from data source
  `/home/sarmiento/Determining-Best-EEZs-For-Oysters/data/wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```
# bathymetry raster
bathymetry <- rast("data/depth.tif")

# SST rasters
SST2008 <- rast("data/average_annual_sst_2008.tif")
SST2009 <- rast("data/average_annual_sst_2009.tif")
SST2010 <- rast("data/average_annual_sst_2010.tif")
SST2011 <- rast("data/average_annual_sst_2011.tif")
SST2012 <- rast("data/average_annual_sst_2012.tif")
SST <- c(SST2008, SST2009, SST2010, SST2011, SST2012)
```

Check for CRS match

```
st_crs(SST) == st_crs(SST2010)
```

```
[1] TRUE
```

```
st_crs(bathymetry) == st_crs(SST)
```

```
[1] TRUE
```

```
st_crs(bathymetry) == st_crs(EEZ)
```

```
[1] TRUE
```

Process data

```
# Mean sst
mean_sst <- mean(SST, na.rm = TRUE)
mean_sst <- mean_sst - 273.15

#Crop depth bathymetry
cropped_depth <- resample(bathymetry, y = mean_sst, method = "bilinear")
cropped_depth <- project(cropped_depth, mean_sst) # match the resolution of SST

# Full match of Depth and SST
stack <- c(mean_sst, cropped_depth)
```

Find suitable locations

```
# Ideal SST between 11-30°C
sst_mask <- mean_sst
sst_mask[sst_mask < 11] <- 0
sst_mask[sst_mask >= 11] <- 1
sst_mask[sst_mask > 30] <- 0

# Ideal depth between 0-70 meters
depth_mask <- cropped_depth
depth_mask[depth_mask >= -70] <- 1
depth_mask[depth_mask < -70] <- 0

# Combine depth and sst mask
sst_and_depth <- c(sst_mask, depth_mask)
names(sst_and_depth) <- c("sst", "depth")

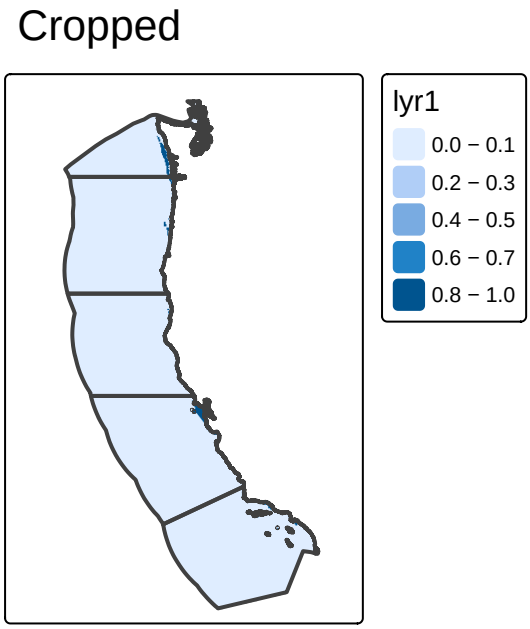
multiply_func <- function(sst, depth) {
  sst*depth
}
```

```r
suitable_raster <- lapp(sst_and_depth, fun = multiply_func)
```

Crop suitable areas to EEZ areas

```r
suitable_raster_cropped <- crop(suitable_raster, EEZ)
suitable_raster_masked <- mask(suitable_raster, EEZ)

tm_shape(suitable_raster_masked) +
  tm_raster() +
tm_shape(EEZ) +
  tm_borders(lwd = 2) +
tm_title(text = "Cropped")
```



Determine the most suitable EEZ

Find area of grid cells

```
#Changes EEZ from vector into raster
eez_raster <- as.factor(rasterize(EEZ, suitable_raster, field = "rgn_id"))

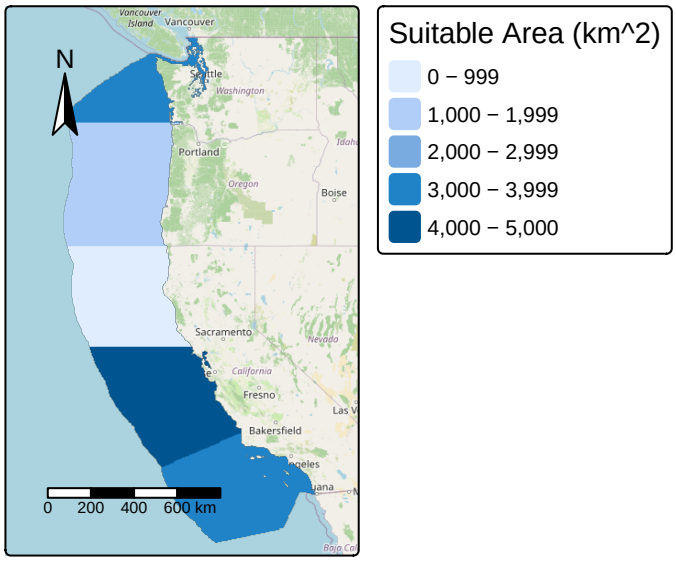
area <- cellSize(eez_raster, unit = "km")

suitable_area <- lapp(c(area, suitable_raster), fun = function(a,s) a*s)
suitable_area_eez <- zonal(suitable_area, eez_raster, fun = 'sum', na.rm = TRUE, as.raster =
names(suitable_area_eez) <- "Suitable Area (km^2)"
```

Final map

```
tm_shape(suitable_area_eez) +
  tm_raster() +
  tm_basemap('OpenStreetMap') +
  tm_title(glue("Oyster Suitable Area Amounts in EEZ zones")) +
  tm_scalebar(position = c("left", "bottom")) +
  tm_compass(position = c("left", "top"))
```

Oyster Suitable Area Amounts in EEZ zones



This map shows that the best EEZ for oyster cultivation is around the coast from Sacramento to Bakersfield. Darker blue indicates higher suitability for oyster cultivation based on temperature and depth.

Function for chosen species

Chosen species: Pacific oyster Ideal range: 0 - 40 meters, −1.8 to 35 °C.

```
OysterSuitabiliyFunc <- function(TempMin, TempMax, DepthMin, DepthMax, Species) {
  #Find suitable locations
  # Ideal SST between TempMin and TempMax
sst_mask <- mean_sst
sst_mask[sst_mask < TempMin] <- 0
sst_mask[sst_mask >= TempMin] <- 1
sst_mask[sst_mask > TempMax] <- 0

# Ideal depth between DepthMin and DepthMax
depth_mask <- cropped_depth
depth_mask[depth_mask > DepthMin] <- 0
depth_mask[depth_mask <= DepthMin] <- 1
depth_mask[depth_mask < DepthMax] <- 0
```

```r
# Combine depth and sst mask
sst_and_depth <- c(sst_mask, depth_mask)
names(sst_and_depth) <- c("sst", "depth")

multiply_func <- function(sst, depth) {
  sst*depth
}

suitable_raster <- lapp(sst_and_depth, fun = multiply_func)

# Crop suitable areas to EEZ areas
suitable_raster_cropped <- crop(suitable_raster, EEZ)
suitable_raster_masked <- mask(suitable_raster, EEZ)

#Changes EEZ from vector into raster
eez_raster <- as.factor(rasterize(EEZ, suitable_raster, field = "rgn_id"))

# Find area for each EEZ
area <- cellSize(eez_raster, unit = "km")

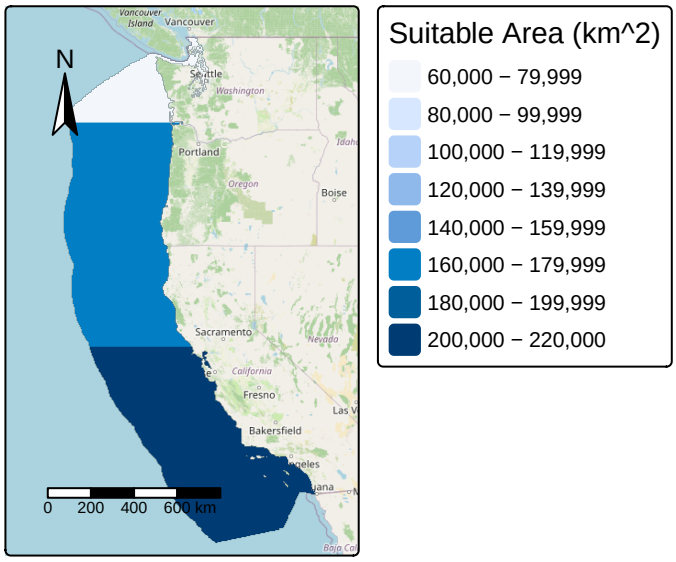
suitable_area <- lapp(c(area, suitable_raster), fun = function(a,s) a*s)
suitable_area_eez <- zonal(suitable_area, eez_raster, fun = 'sum', na.rm = TRUE, as.raster =
names(suitable_area_eez) <- "Suitable Area (km^2)"

#Final map creation
tm_shape(suitable_area_eez) +
  tm_raster() +
  tm_basemap('OpenStreetMap') +
  tm_title(glue("Suitable Area Amounts in EEZ's for {Species}")) +
  tm_scalebar(position = c("left", "bottom")) +
  tm_compass(position = c("left", "top"))
}
```

Running function for the Pacific Oyster

```r
OysterSuitabiliyFunc(-1.8, 35, 0, -40, "Pacific oyster")
```

Suitable Area Amounts in EEZ's for Pacific oyster



For the Pacific Oyster's ideal range of 0 - 40 meters and temperature range of −1.8 to 35 °C, we can see that best place for cultivation seems to be from the Sacramento coast all the way down to San Diego.